

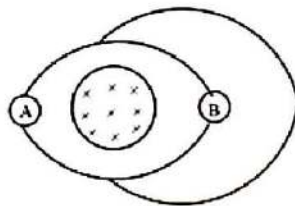


MAGNETISM_EMI & AC

Section-I

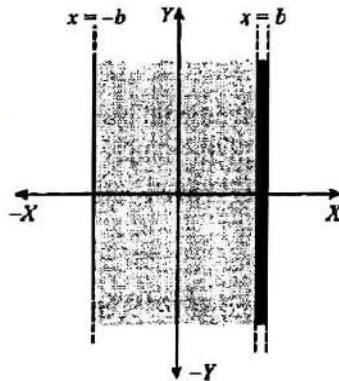
(Multiple Correct Choice Type)

2. Which of the following statement(s) is/are correct (Neglect interaction between two particles):
- A) A charged particle enters a region of uniform magnetic field at an angle 85° to magnetic lines of force. The path of the particle is a circle.
- B) An electron and proton are moving with the same kinetic energy along the same direction. When they pass through uniform magnetic field perpendicular to their direction of motion, they describe circular path.
- C) There is no change in the energy of a charged particle moving in a magnetic field although magnetic force acts on it.
- D) Two electrons entering with the same speed but in opposite direction in a uniform transverse magnetic field move in same sense of rotation with equal radius
28. A very long solenoid perpendicular to the page generates a magnetic field in to the page whose magnitude increases with time. Two bulb A and B connected by a wire is glowing due to induced current in loop. Now two points diametrically opposite are shorted with another wire lying to the right of bulb in the plane of page. After the shortening wire is inserted

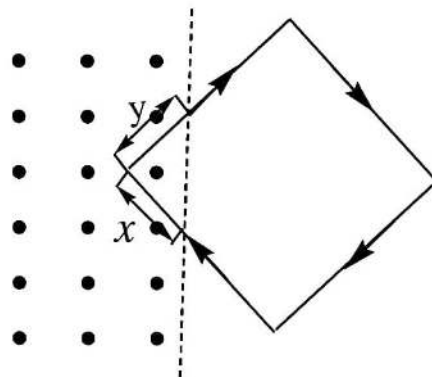


- A) Bulb A goes out (Turns off) B) Bulb A gets dimmer
C) Bulb B goes out (Turns off) D) Bulb A gets brighter

3. The figure contains an infinite slab having current per unit area of $\vec{J}_1 = J\hat{k}$ between the infinite planes at $x = -b$ and $x = b$. Slightly to the right of $x = b$, an infinite thin sheet is kept. It carries a current per unit length $\vec{J}_2 = 2bJ(-\hat{k})$. Choose the correct option(s) :

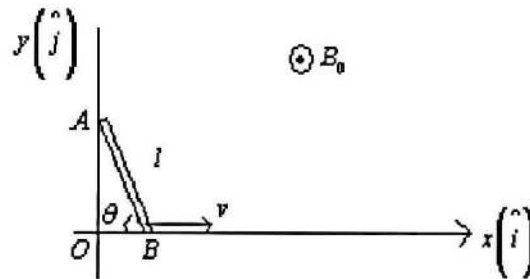


- A) The magnitude of magnetic field due to sheet at any point outside it is $\sim_0 Jb$
- B) The magnitude of magnetic field due to sheet at any point is $\frac{\sim_0 Jb}{2}$
- C) The magnitude of magnetic field due to slab at a point inside the slab must be independent of y co-ordinate and y component of field must be a function of x
- D) Magnitude of magnetic field at a point (x, y) inside slab, due to slab is $\sim_0 Jx$
6. On a frictionless table-top, a rigid square loop of side length l carrying an electric current is placed. A magnetic field is set as shown and then the loop is released. Considering the different lengths of the side segments x and y ($x < y$) shown in the figure which of the following conclusions can be made? (plane of the loop lies on the plane of the table)



- A) If $x < \frac{l}{2}$ and $y = \frac{l}{2}$, the loop starts rotating anticlockwise
- B) If $x < \frac{l}{2}$ and $y < \frac{l}{2}$, the loop starts rotating anti-clockwise
- C) If $x < \frac{l}{2}$ and $y > \frac{l}{2}$, the loop starts rotating clockwise
- D) If $x < \frac{l}{2}$ and $y > \frac{l}{2}$, the loop starts rotating in counter clock wise direction

29. A thin conducting rod of length l is moved such that its end B moves along the x-axis while end A moves along the y – axis. A uniform magnetic field $B = B_0 \hat{k}$ exists in the region. At some instant, velocity of end B is v and the rod makes an angle of θ with the x-axis as shown in the figure. Then at this instant



- A) Angular speed of rod AB is $\omega = \frac{V}{L \sin \theta}$
- B) Angular speed of rod AB is $\omega = \frac{V}{L \cos \theta}$
- C) emf induced in rod AB is $\frac{B \ell v \sin 2\theta}{2 \cos \theta}$
- D) emf induced in rod AB is $\frac{B \ell v \cos 2\theta}{2 \sin \theta}$
30. A thin non-conducting ring of mass m , radius 'a' carrying a charge q can rotate freely about its own axis which is vertical. At the initial moment the ring was at rest and no magnetic field was present. At instant $t = 0$, a uniform magnetic field is switched on which is vertically downwards and increases with time according to the law $B = B_0 \sin t$. Neglecting magnetism induced due to rotational motion of the ring, which of the following statements is/are correct:

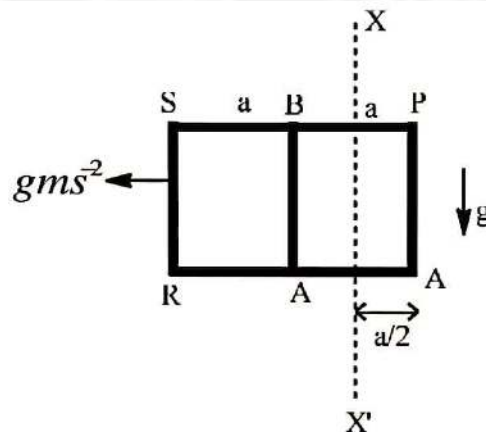
A) angular acceleration of the ring is $\frac{qB_0}{2m} \cos t$

B) angular acceleration of the ring is $\frac{qB_0}{m} \sin t$

C) power developed by forces on the ring as a function of time is $\frac{q^2 B_0^2 a^2 \sin 2t}{8m}$

D) power developed by forces on the ring as a function of time is $\left(\frac{q^2 B_0^2 a^2 \cos^2 t}{4m} \right) t$

27. A conducting loop is shown figure, it is made of seven identical wires each of length a , resistivity ρ , area of cross-section S_0 . The frame lies in a vertical plane. XX' is an imaginary boundary separating space in two parts. Right to XX' , a uniform gravitational field $\vec{g}$ exists (figure) whereas no gravitational field is present left of XX' . Instantaneously frame is accelerating in leftwards with acceleration gm/s^2 . e and m is electronic charge and mass electron respectively. Ignore inductive and capacitive properties of loop. Instantaneously



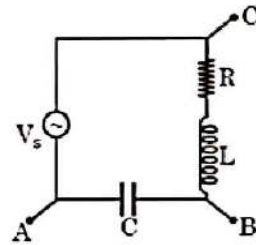
A) The magnitude of potential difference between points P and Q is $\frac{7mga}{15e}$

B) The magnitude of potential difference between points P and Q is $\frac{8}{15} \frac{mga}{e}$

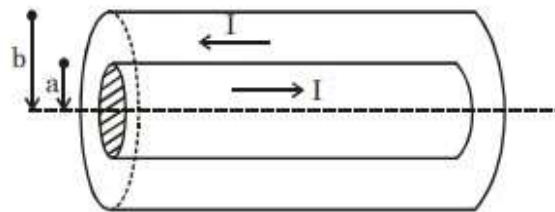
C) Current in branch RS is $\frac{mgS_0}{15\rho e}$

D) Current in branch RS is $\frac{4mgS_0}{15\rho e}$

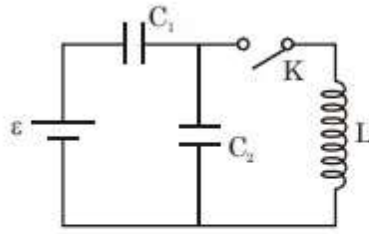
31. Consider a series LRC circuit with an ac voltage V_s of varying frequency ω , as shown in the figure. A circuit which allows low frequency voltages, but blocks high frequency voltages is called low pass filter (LPF) and circuit which allows high frequency voltages but blocks low frequency voltages is called High-pass filter (HPF). When the output voltage (V_0) is measured :-



- A) across A and B, the circuit behaves like a LPF because $\lim_{\omega \rightarrow 0} \frac{V_0}{V_s} = 1.0$ and $\lim_{\omega \rightarrow \infty} \frac{V_0}{V_s} = 0$.
- B) across B and C, the circuit behaves like a HPF because $\lim_{\omega \rightarrow 0} \frac{V_0}{V_s} = 0$ and $\lim_{\omega \rightarrow \infty} \frac{V_0}{V_s} = 1.0$.
- C) across B and C, the circuit behaves like a HPF because $\lim_{\omega \rightarrow 0} \frac{V_0}{V_s} = 0$ and $\lim_{\omega \rightarrow \infty} \frac{V_0}{V_s} = 1.0$.
- D) across A and B, the circuit behaves like a LPF because $\lim_{\omega \rightarrow 0} \frac{V_0}{V_s} = 0$ and $\lim_{\omega \rightarrow \infty} \frac{V_0}{V_s} = 1.0$.
12. A long co-axial cable carries current I as shown in figure. (Current flows down the surface of inner cylinder of radius a and back along the outer cylinder of radius b .) Choose the correct statement(s).



- (A) The magnetic energy stored in a section of length ℓ is $\frac{\mu_0 I^2 \ell}{4\pi} \ln\left(\frac{b}{a}\right)$
- (B) The magnetic field between the cylinders have magnitude of $\frac{\mu_0 I}{2\pi(b-a)}$
- (C) Self inductance of the cable of length ℓ is $\frac{\mu_0 \ell}{4\pi} \ln\left(\frac{b}{a}\right)$
- (D) Magnetic field outside the cylinder is zero.
13. Consider the electrical circuit given below; initially chargeless capacitors C_1 and C_2 were connected to a battery and at certain moment, the key K will be closed. After that moment, current and voltage will start oscillating. For these oscillations :



- (A) When charge on C_2 is maximum, current through C_1 is same as current in L .
 (B) When charge on C_1 is maximum, power delivered by battery is positive.
 (C) Charge on C_2 is maximum when current through L is maximum.
 (D) None of these

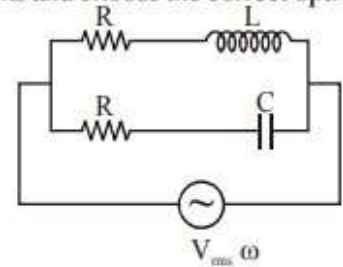
14. In this circuit given, the value of $R = \sqrt{\frac{L}{C}}$. Consider the following statements and choose the correct option.

(i) RMS value of current through a.c. source is $\frac{V_{rms}}{2R}$.

(ii) Impedance of circuit is independent of ω .

(iii) Impedance of circuit depends upon ω .

(iv) Current through the source is $\frac{V_{rms}}{\sqrt{R^2 + \omega^2 L^2}}$.



(A) only (iii) and (iv) are correct

(B) only (ii) is correct

(C) only (i) and (ii) are correct

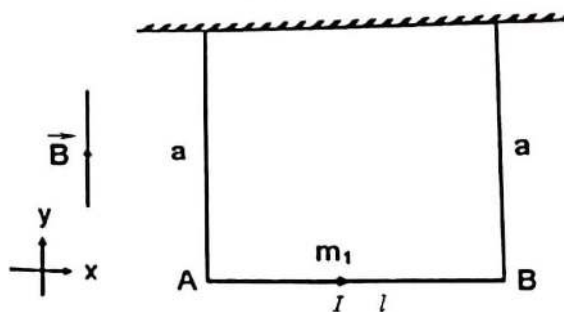
(D) only (iv) is correct

Section-II

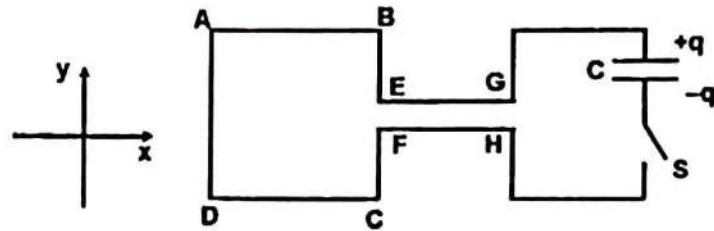
(Integer Type)

10. A conducting rod AB of mass m and length l is supported by conducting massless wires and carrying a steady current I as shown in the figure. There exist a uniform magnetic field B in the vertically upward direction. The rod is slightly displaced into the paper such that rod always remains parallel to x -axis and released to perform SHM. If its time

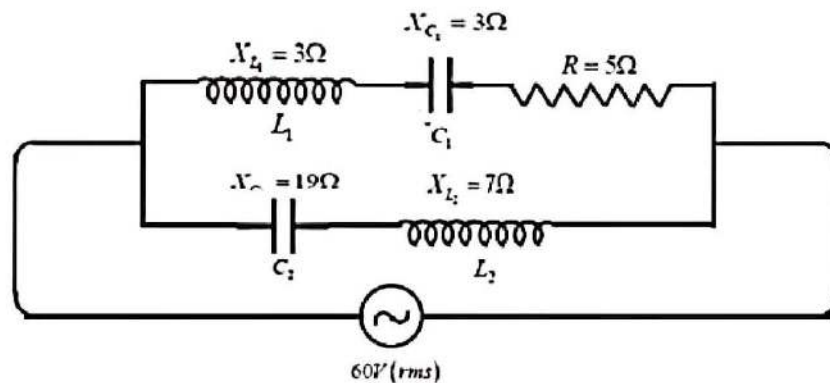
period is $2\pi \sqrt{\frac{ka}{g} \left[1 + \left(\frac{BIl}{mg} \right)^2 \right]^{-1/4}}$ find the value of k . (neglect motional emf)



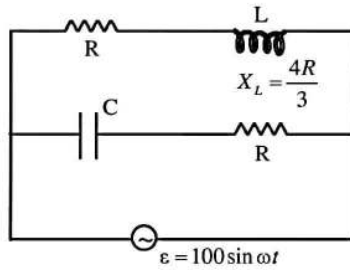
11. A straight cylindrical conductor of length l and radius 'a' has resistivity $\rho = kr^n$ where r is radial distance from its axis. Flat ends of the conductor are maintained at a potential difference V . For points within the conductor ($r < a$), the value of 'n' for which the magnitude of magnetic field inside the conductor remains constant is_____
12. A conducting square loop of side 'a' is kept on a sufficient rough horizontal surface and connected to a charged capacitor as shown in the figure. The arms AB, BE, FC, CD and DA are rigid but connecting wires EG, FH can behave as strings, so that the whole loop can rotate about BC. The mass of each arm is m and the gap between points E and F can be neglected. A uniform magnetic field is present in the x direction. Upon closing the switch $\sqrt{\frac{5km^2g}{3B^2a}}$ is the amount of charge flown in circuit so as to rotate the loop by 90° about BC, then find the value of k . (neglect the resistance of wires)



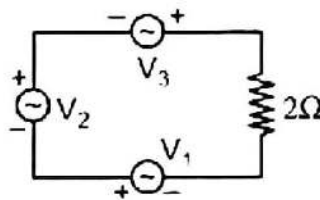
22. In the given A.C circuit the r.m.s voltage of the source is 60V then the r.m.s current passing through the source is $(10+x)$ A then the value of x is



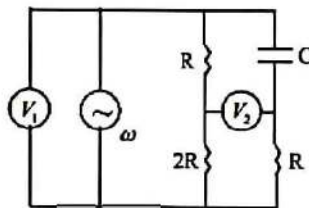
23. In the AC circuit shown in figure the capacitance of the capacitor is chosen in such a way that the average power P delivered by the source to the circuit is maximum. If $R = 10\Omega$ then the values of P is $N \times 100W$. Where N is



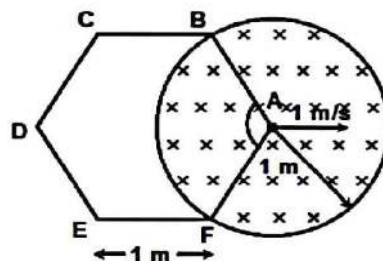
21. Three alternating voltage sources $V_1 = 3\sin\omega t$ volt, $V_2 = 5\sin(\omega t + \phi_1)$ volt and $V_3 = 5\sin(\omega t - \phi_2)$ volt connected across a resistance $R = 2\Omega$ as shown in the figure (where ϕ_1 and ϕ_2 corresponds to 60° and 60° respectively). Find the peak current (in amp.) through the resistor.



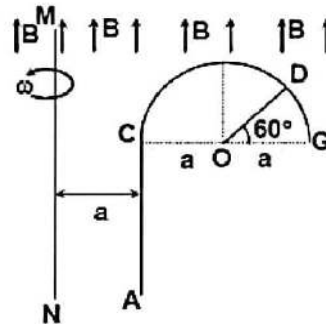
24. A circuit shown consists of three resistors, one capacitor, 2 ideal AC voltmeters and an alternating voltage source of angular frequency ω . Take $R\omega C = 1$. If the reading of V_1 is V , then the reading of V_2 is $\frac{V}{3}\sqrt{\frac{n}{2}}$, then n is



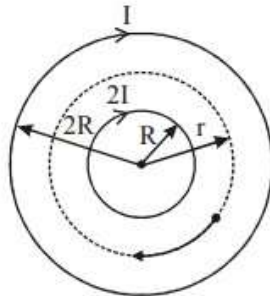
25. A cylindrical region of radius 1m has instantaneous homogenous magnetic field of $\sqrt{3}$ T and it is increasing at a rate of $\frac{6}{\pi}$ T/s. A regular hexagonal loop ABCDEFA of side 1m is being drawn in to the region with a constant speed of 1 m/s as shown in the figure. What is the magnitude of emf developed in the loop just after the shown instant when the corner A of the hexagon is coinciding with the centre of the circle?



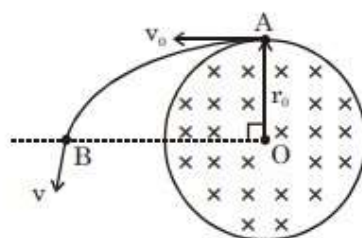
26. A conductor ACDG consists of a straight segment AC $\equiv$ 0.5m and a semi-circular arc CDG of radius $a = 20\text{cm}$. A uniform vertical magnetic field $B = 5\text{ Tesla}$ is existing in the region. Now the conductor is rotated with a constant angular velocity $\omega = 10\text{ rad/s}$ about a vertical axis MN parallel to the magnetic field as shown. Then find the emf (in volt) induced between the points A and D of the conductor as shown.



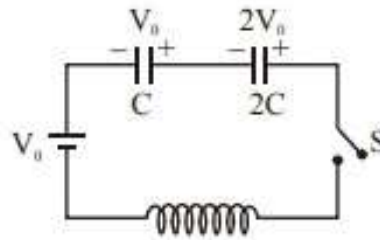
1. A long solenoid contains another coaxial solenoid (whose radius R is half of its own). Their coils have the same number of turns per unit length and initially both carry no current. At the same instant current starts increasing linearly with time in both solenoids. At any moment the current flowing in the inner coil is twice as large as that in the outer one and their directions are the same. As a result of the increasing currents a charged particle, initially at rest between the solenoids, starts moving along a circular trajectory of radius r (see figure). The value of $\frac{r\sqrt{2}}{R}$ is.



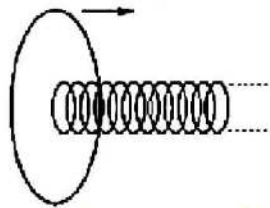
2. Consider a cylindrical region of uniform but time varying magnetic field. Now consider a plane perpendicular to this field. Top view of this cylindrical magnetic field on the plane is shown in the diagram which is a circular region of radius $r_0 = 1\text{m}$. A particle of mass $p\text{ kg}$ & charge 2C is projected with velocity of $v_0 = \sqrt{5}\text{ m/s}$, tangential to this circle from point A. This particle intersects another radial line OB at point B with velocity v . If the magnetic field is increasing at the rate of 4 T/s then find the value of v .



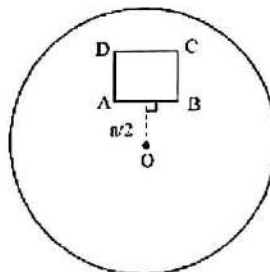
3. Initially current through inductor is zero. After closing of switch when current through inductor becomes maximum, the charge on capacitor C is $\frac{nCV_0}{3}$. Find the value of n.



19. A conducting loop having a radius $2a$ is held co-axially near the end of a tightly wound semi-infinite solenoid carrying a current i and having 'n' number of turns per unit length. The radius of the solenoid is a . Now the loop is moved co-axially somewhere towards the middle of the solenoid as shown. If the resistance of the loop is R , the amount of the charge that flows through the loop in the process is $\frac{\mu_0 n i \pi a^2}{KR}$ then K is



20. Figure shows cross-section view of a solenoid. There exists uniform time dependent magnetic field $\beta_t = At^2 + Bt$ along the axis of solenoid. A square ($a \times a$) conducting frame is kept in solenoid keeping its plane perpendicular to axis of solenoid. The side of square closest to axis of solenoid is at a distance $a/2$ from axis. Find ratio of magnitude of potential difference in branch CD to AB? Resistance per unit length of square frame is uniform.

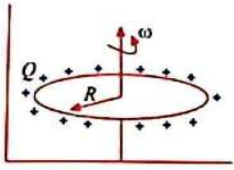
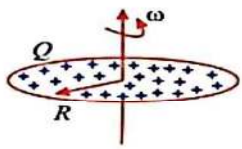
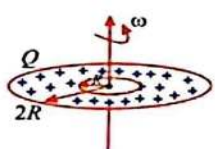
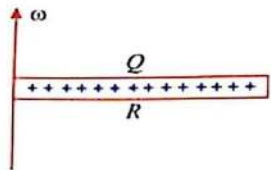


Section-III

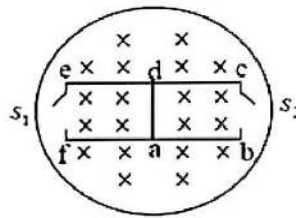
(Matching)

Answer Q,13, Q,14 and Q,15 by appropriately matching the information given in the three columns of the following table.

A charge Q is distributed uniformly on the different non conductors as shown in the column I. Each object rotates with uniform angular velocity $\vec{\omega}$ about the axis shown in the column – I. The magnetic field at the centre (i.e. on the axis of rotation) and magnetic moment are given in column – II and column – III. Match the column – I with column – II and column – III.

	Column- I		Column-II		Column- III
1)	A ring rotating about its own axis 	(P)	$B = \frac{\mu_0 Q \omega}{6f R}$	(A)	$M = \frac{Q \omega R^2}{4}$
2)	A disc rotating about its own axis 	(Q)	$B = \frac{\mu_0 Q \omega}{24f R}$	(B)	$M = \frac{15 Q \omega R^2}{4}$
3)	An annular disc of inner radius R and outer radius 2R rotating about its own axis 	(R)	$B = \frac{\mu_0 Q \omega}{4f R}$	(C)	$M = \frac{Q \omega R^2}{6}$
4)	A linear thin rod rotating about an axis perpendicular to axis and passing through one end 	(S)	$B = \frac{\mu_0 Q \omega}{2f R}$	(D)	$M = \frac{Q \omega R^2}{2}$

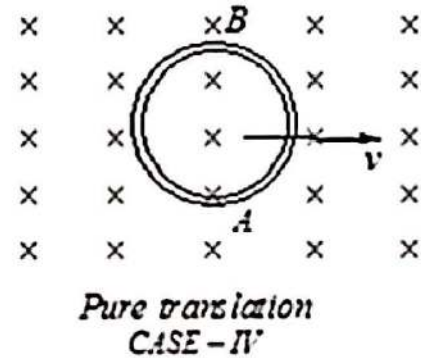
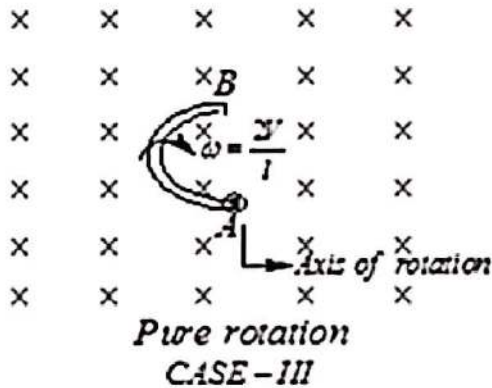
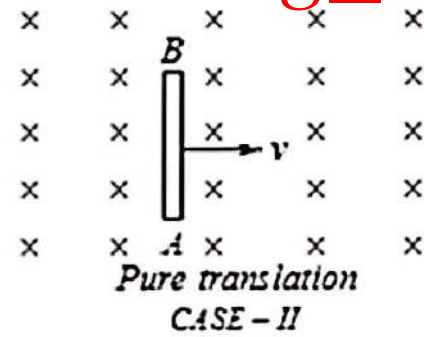
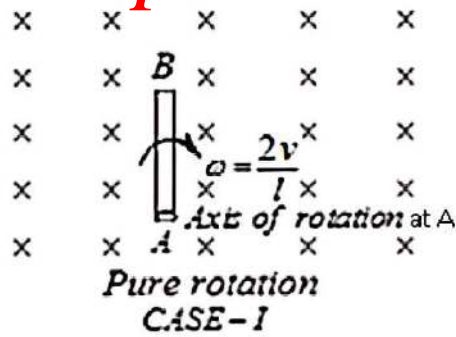
13. Which of the following is correct ?
 A) 1, R, D B) 1, S, B C) 1, R, A D) 1, Q, A
14. Which of the following is correct ?
 A) 2, S, C B) 3, Q, A C) 2, S, A D) 4, Q, A
15. Which of the following is correct?
 A) 3, Q, A B) 3, S, B C) 2, S, C D) 1, S, A
35. The magnetic field in the cylindrical region shown in figure increases at a constant rate of 10.0 mT/s . Each side of the square loop abcd and defa has a length of 2.00 cm and resistance of 2.00Ω . Correctly match the current in the wire ad along with its direction in four different situations as listed in column-I with the values given in column-II.



Column – I		Column – II	
A)	The switch S_1 is closed but S_2 is open	P)	$5 \times 10^{-7} \text{ A}$, d to a
B)	S_1 is open but S_2 is closed	Q)	$5 \times 10^{-7} \text{ A}$, a to d
C)	Both S_1 and S_2 are open	R)	$2.5 \times 10^{-8} \text{ A}$, d to a
D)	Both S_1 and S_2 are closed	S)	No current flows

- A) A – Q ; B – Q ; C – S ; D – R B) A – P ; B – Q ; C – S ; D – S
 C) A – S ; B – Q ; C – S ; D – P D) A – Q ; B – P ; C – S ; D – S

36. Consider four cases in which conducting wire is moved with constant velocity v or rotated with constant angular velocity ω in uniform and constant magnetic field B_0 , which is same for all cases and extended upto infinity. Distance between points A and B directly in each case is l . In case of pure rotation plane of rotation is perpendicular to direction of magnetic field. Potential difference between points B and A is $V_B - V_A$.



Match the following:

	Column I		Column II
(A)	$ V_B - V_A = B_0 l v$	(p)	Case I
(B)	Rate of area swept by the conductor $= \frac{dA}{dt} = vl$	(q)	Case II
(C)	Uniform electric field in the conductor	(r)	Case III
(D)	When velocity or angular velocity remains constant, then the external force required on the conductor is zero	(s)	Case IV

- A) A - PQRS ; B - PQR ; C - QS ; D - QS
 B) A - PQRS ; B - PQRS ; C - QS ; D - PQRS
 C) A - PQRS ; B - PQRS ; C - QS ; D - QS
 D) A - PQRS ; B - PQRS ; C - PQR ; D - PQRS

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KEY:

1	BCD	2	CD	3	ACD	4	AB	5	AD
6	AC	7	BC	8	AC	9	AD	10	A
11	B	12	1	13	1	14	4 OR 5	15	3
16	6.8	17	4	18	5	19	5	20	5.25
21	2	22	3	23	5	24	2	25	1
26	A	27	C	28	B	29	D	30	B

